

Detecting Suspicious Activities in Waste Transport Data via Temporal and Network-Aware Change Detection

Introduction

Waste management plays a critical role in **environmental protection and public health**, but it is also vulnerable to **fraud, non-compliance, and organized criminal activities**. In Portugal, authorities such as IGAMAOT perform inspections to detect these issues, yet limited resources demand an efficient, targeted approach for prioritizing inspections. This thesis addresses the challenge of developing an **anomaly detection framework** for the Portuguese waste management network, using anonymized 2022 waste transport data provided by the Portuguese Environmental Agency (APA).

On one hand, traditional detection methods fall short in capturing the **temporal irregularity and relational nature of these anomalies**. To overcome this, we propose a novel anomaly detection framework that combines statistical change detection techniques (**Page-Hinkley** and **CUSUM** tests) with a deep learning approach (**LSTM-VAE**). For these methods, we develop a set of **novel network and operational features** indicative of suspicious behavior. By analyzing both the temporal evolution and network structure of company behavior, this framework aims to uncover **sudden deviations or unusual patterns** indicative of illicit activity.

Data

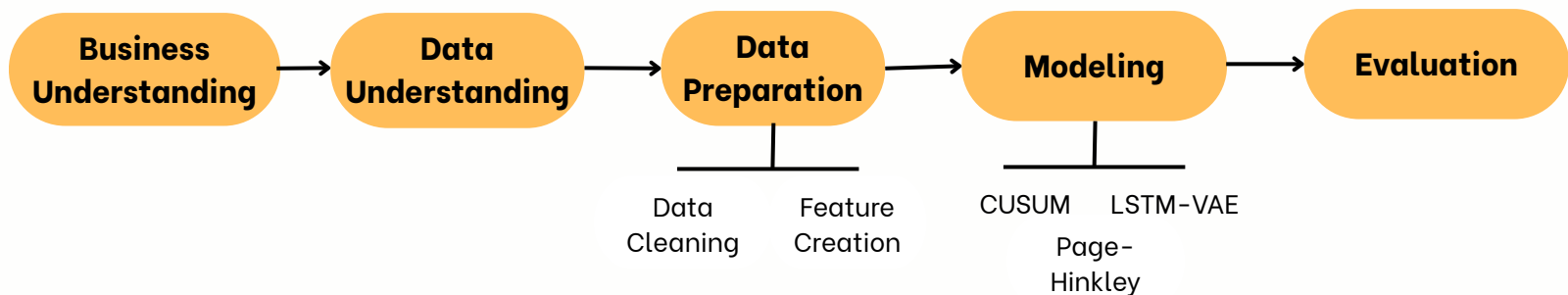
Over **4.5 million waste transport records** taking place in **Portugal in 2022** indicating type and quantity of waste, date of transfer, and identification of waste senders and receivers.

From this data a **dynamic network was constructed**, to develop the network-based temporal features. The **target** of our anomaly detection were the 1 591 unique **waste receiving companies**.

Four of these companies are **confirmed to be anomalous** and will thus be used for evaluation purposes.

Methodology

The methodology employed was based on the CRISP-DM framework, dividing the work into distinct, well defined phases:



Feature creation

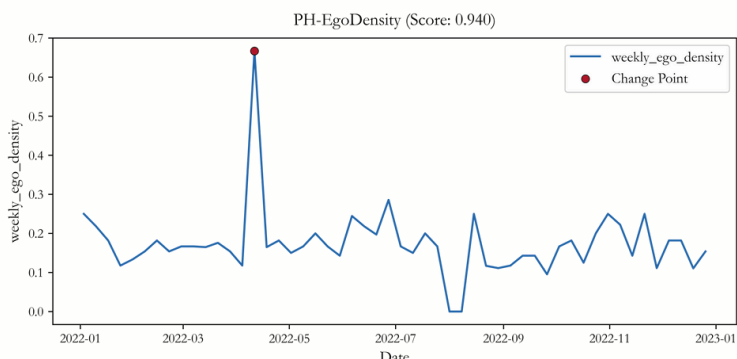
The following time series features were created for all waste receiving companies, in weekly or monthly periodicity:

- Connectivity with individual waste senders in triangles
- Betweenness centrality
- Ego network density
- Waste quantity received (total and from individual senders)
- Number of triangles
- First component of PCA of 5 activity features

Page-Hinkley and CUSUM

These change detection techniques were applied to **detect significant deviations in the mean of the timeseries features** for each company. Significant deviations in the network behavior can signal suspicious activities, as these are often sporadic rather than continuous events. The tests were applied for each company on previously standardised features: connectivity with senders in triangles, ego network density and the first component of PCA.

The company **anomaly score** was defined as the **sum of the top 3 detected change strengths**.



LSTM-VAE

The Long Short-term Memory Variational Autoencoder was implemented to **learn complex, non-linear temporal patterns** in two sets of features:

- Betweenness centrality and waste quantity received (**LSTM-VAE-2F**)
- Connectivity with senders in triangles, betweenness centrality, and quantity received from individuals (**LSTM-VAE-3F**)

The model was trained on the standardized companies' timeseries to minimize a combined reconstruction loss and a Kullback-Leibler divergence term.

The **anomaly scores** were obtained as the **mean of the 10 highest normalized reconstruction errors**, to detect companies with periods where their behavior was particularly difficult to reconstruct and therefore suspicious.

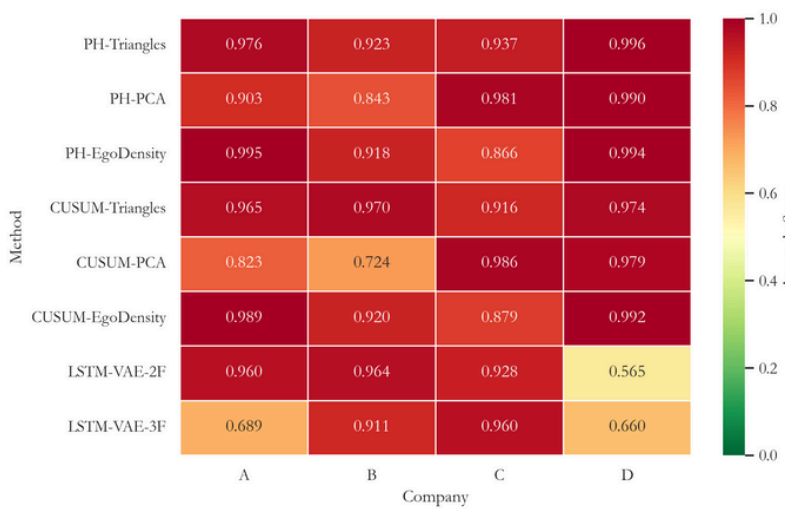
The models were run 100 times with different random initializations.

Evaluation

The methods were evaluated based on the set of **4 companies labeled as anomalous**. All companies consistently showed a **very high score in at least six of the methods**.

The change detection techniques, particularly those based on triangle features, consistently identified the companies with high scores (above 0.9).

The LSTM-VAE models showed a limitation by failing to detect Company D.



Anomaly Scoring

To combine the different methods and obtain a final anomaly score, five methods were selected:

- CUSUM on connectivity with senders in triangles (CUSUM-Triangles)
- PH on ego network density (PH-EgoDensity)
- PH on the first component of PCA (PH-PCA)
- LSTM-VAE-2F
- LSTM-VAE-3F

The companies were ranked for each method, and the **final anomaly score** was defined as the **median standardized rank**. The median was selected due to the **complementarity of the approaches**: to obtain a high score a company must obtain a high score in at least 3 of the methods, but may fail in the remaining 2.

The 5 methods presented **low, although positive, correlation** among each other (between 0.13 and 0.34).

All labeled anomalous companies were detected by at least 3 of the 5 methods, and they were in the **top 2.67% of the anomaly scores**.

Company	Final Score	Percentile of final score
A	0.960	99.81
B	0.918	98.33
C	0.928	98.81
D	0.974	99.94

Conclusion

This thesis successfully developed and validated a framework for detecting suspicious activities within waste transport networks, combining change detection techniques with LSTM-VAE models. The approach ensured that all labeled anomalous companies were detected by at least three of the five techniques used, and that these companies were placed in the top 2.67% of anomaly scores. The five methods obtained low, although positive correlation (between 0.13 and 0.34) in the anomaly scores, suggesting the **complementarity of the approaches** in detecting a wider range of anomalies.

These results validate the methodology's potential for building a **scalable analytical tool** to monitor large scale waste transport records, identify companies with suspicious behaviors and thus **support regulatory authorities** in prioritizing company inspections in the waste management network. However, a larger set of labeled companies is necessary for further model validation. Additionally, the inclusion of transport records from 2023 and 2024, for more robust modeling of the temporal patterns, and a refinement of the deep learning approach would further improve the proposed framework.